

Agent Guardrail Recommendations

Guardrail Recommendations — IIoT Manufacturing Agentic AI Platform

Based on the 12 agents defined in the uploaded platform simulation, here are tailored guardrails for each, mapped to the 9 standard categories. Existing platform guardrails (from the Admin Console — are noted where relevant, with additions/refinements suggested.

1. Operations Agent

*Monitors throughput/OEE, recommends corrective scheduling —
owner: Plant Digital Head*

Category	Suggested Guardrail
Input validation	Sanity-check MES/SCADA feed values against physical bounds (e.g., OEE can't exceed 100%, negative throughput rejected) before they drive a recommendation
Output filtering	Recommendations must cite the specific line/shift data point that triggered them — no generic or unexplained scheduling advice
Tool/action permissions	Agent may read SAP production orders and draft schedule changes; it cannot commit a schedule change directly in SAP
Human-in-the-loop	Shift Supervisor or Plant Head must confirm any corrective scheduling change before it takes effect (already noted in KB)
Scope/topic boundaries	Restricted to throughput/OEE/scheduling; must not make staffing, safety, or financial recommendations
Rate limits / spend caps	Cap number of scheduling recommendations per shift (e.g., max 5) to avoid alert fatigue/thrashing
Output structure validation	Recommendation payload must match a fixed schema (line ID, metric, deviation %, proposed action) before reaching the supervisor UI
Self-monitoring	Agent checks its own recommendation against the last 3 it made for the same line — flag if contradicting itself
Fallback/escalation	If no supervisor response in 15 min, auto-escalate to Plant Head

(already implemented)

2. Maintenance Agent

LSTM predictive failure model, auto-raises work orders

Category	Suggested Guardrail
Input validation	Reject vibration/temperature sensor readings outside plausible physical ranges (sensor fault vs. real signal)
Output filtering	Failure prediction must include confidence score; suppress low-confidence (<70%) alerts from auto-triggering a work order
Tool/action permissions	Can create a work order in SAP/CMMS automatically; cannot approve or dispatch the repair action itself — that stays human-approved
Human-in-the-loop	Repair action (cost/downtime impact) requires Maintenance Lead sign-off, even though the work order creation is automatic
Scope/topic boundaries	Limited to rotating/monitored equipment with sensor coverage; must not extrapolate failure predictions to unmonitored assets
Rate limits / spend caps	Cap auto-generated work orders per asset per week to prevent duplicate/noisy tickets
Output structure validation	Work order must populate required SAP fields (asset ID, predicted failure mode, lead time) or it's rejected before submission
Self-monitoring	Track prediction accuracy over time (false positive rate); flag model drift for review
Fallback/escalation	If model confidence is ambiguous, default to "flag for manual inspection" rather than silently taking no action

3. Safety & Quality Agent

AI vision for PPE, intrusion detection, defect inspection

Category	Suggested Guardrail
Input validation	Verify camera feed integrity/timestamp freshness before acting on vision output (reject stale or tampered feeds)
Output filtering	Blur/mask faces and license plates by default on any output shared outside safety review (already an existing platform guardrail) Can flag and notify ; cannot autonomously halt a production

Tool/action permissions	line — that requires supervisor action
Human-in-the-loop	High-risk safety events (e.g., intrusion into a restricted zone) require immediate supervisor acknowledgment, not just a passive log
Scope/topic boundaries	Vision analysis restricted to PPE/safety/quality use cases — not repurposed for general employee monitoring or productivity tracking
Rate limits / spend caps	Cap redundant alerts for the same detected event within a short window (avoid alert storms from continuous frames)
Output structure validation	Detection events must include zone ID, timestamp, confidence, and image reference in a fixed schema for downstream logging
Self-monitoring	Periodic self-check against known false-positive patterns (e.g., mannequins, reflections) before escalating
Fallback/escalation	If vision confidence is borderline, route to human review queue rather than auto-classifying as violation or non-violation

4. Energy Agent

kWh tracking, peak-demand forecasting/shaving

Category	Suggested Guardrail
Input validation	Validate smart-meter readings against expected load ranges per line before triggering demand-shaving logic
Output filtering	Forecasts must state assumptions (season, production plan) alongside the number — no bare predictions
Tool/action permissions	Can recommend load shifts; cannot directly control machine power states without Plant Digital Head sign-off (matches existing KB)
Human-in-the-loop	Any peak-shaving action affecting production lines requires explicit approval before execution
Scope/topic boundaries	Confined to energy/demand topics; should not recommend production schedule changes (that's Operations Agent's domain)
Rate limits / spend caps	Limit number of load-shift recommendations per day to avoid production disruption
Output structure validation	Forecast output must include line ID, time window, and projected kWh in a consistent

	schema for the dashboard
Self-monitoring	Compare forecast vs. actual consumption weekly; flag if forecast error exceeds a set threshold
Fallback/escalation	If sensor data is missing for a line, exclude it from aggregate forecast rather than interpolating silently

5. ESG & Compliance Agent

SO₂/CO₂/permit threshold tracking, auto-generates regulatory reports

Category	Suggested Guardrail
Input validation	Cross-check sensor readings against regulatory reference tables for plausibility before including in a report
Output filtering	Auto-generated regulatory reports must be checked for numerical accuracy and complete required fields before submission-ready status
Tool/action permissions	Can draft and internally route compliance reports; cannot submit directly to a regulator without sign-off
Human-in-the-loop	HSE Officer must review/approve any report before external regulatory submission (high-stakes/legal exposure)
Scope/topic boundaries	Restricted to environmental/permit compliance; should not comment on unrelated safety incidents
Rate limits / spend caps	N/A for actions, but cap redundant “watch item” notifications for the same trend
Output structure validation	Reports must conform to the regulator’s required template/schema — validated before being marked ready
Self-monitoring	Agent re-checks its own threshold logic against the latest regulatory limit table version (limits can change)
Fallback/escalation	If sensor data is incomplete near a filing deadline, escalate to HSE Officer rather than filing with gaps

6. Finance Agent

Real-time P&L, EBITDA, capex-pacing, variance flags

Category	Suggested Guardrail
	Reconcile ERP/invoice inputs against expected ranges; reject

Input validation	clearly erroneous entries (e.g., duplicate invoices) before flagging variance
Output filtering	Variance flags must never expose individual compensation, vendor-sensitive terms, or other confidential financial PII beyond the requester's role
Tool/action permissions	Read-only on ERP/financial data; cannot post journal entries, approve payments, or alter budget lines
Human-in-the-loop	Any budget reallocation suggestion requires Finance Controller approval before action
Scope/topic boundaries	Confined to financial reporting/variance; should not make operational recommendations (e.g., "cut this shift") without Finance Controller framing
Rate limits / spend caps	Cap frequency of variance alerts for the same line item to avoid noise
Output structure validation	Variance report must follow a fixed schema (account, budget, actual, variance %, threshold breached)
Self-monitoring	Cross-validate real-time figures against month-end close reconciliation to catch drift
Fallback/escalation	If ERP data feed is delayed/stale, clearly flag the report as "based on stale data" rather than presenting it as current

7. PPE & Behavior Vision Agent

Vision-based unsafe-act detection (fatigue, restricted-zone entry, collapse)

Category	Suggested Guardrail
Input validation	Confirm camera health/coverage before relying on a zone's feed for a safety-critical detection
Output filtering	Apply PII masking by default on any stored/shared clip; only unmask under formal safety-review process (existing platform guardrail)
Tool/action permissions	Can notify supervisors in real time; cannot trigger physical interventions (e.g., locking down equipment) autonomously
Human-in-the-loop	A detected "collapse" or medical event should trigger an urgent human-verified emergency response path, not just a logged notification
Scope/topic boundaries	Detection limited to defined unsafe-act categories; not to be repurposed for general behavioral surveillance

Rate limits / spend caps	De-duplicate repeated alerts for the same ongoing event (one notification, with updates, not a flood)
Output structure validation	Alerts must include zone, event type, confidence, and timestamp in a consistent schema
Self-monitoring	Track false-positive/negative rates by zone and lighting condition; recalibrate models where accuracy drops
Fallback/escalation	If detection confidence is low for a potential collapse/emergency, err toward escalation (false positive is safer than missed real event)

8. Permit-to-Work Agent

Tracks hot work/confined space/work-at-height permits, auto-escalates expiry

Category	Suggested Guardrail
Input validation	Validate permit records for completeness (required approvals, timestamps) before treating them as “active”
Output filtering	Escalation messages must include permit ID, type, and holder — no ambiguous “a permit expired” notices
Tool/action permissions	Can notify/escalate ; cannot auto-revoke or auto-issue a permit — that remains a human EHS action
Human-in-the-loop	Escalation to HSE Head after 15 min unacknowledged (existing rule) — keep humans as the final approver for permit renewal
Scope/topic boundaries	Confined to permit lifecycle/expiry tracking; not authorized to assess whether underlying work itself is safe to continue
Rate limits / spend caps	Avoid repeat-notification spam for the same expired permit within a short interval
Output structure validation	Escalation payload matches schema: permit ID, type, holder, expiry time, zone
Self-monitoring	Periodically verify its own escalation SLA compliance (did notifications go out within 5 min as designed)
Fallback/escalation	If notification channel (SMS/App) fails, fall back to a secondary channel automatically rather than silently failing

9. Incident & Investigation Agent

Tracks TRIR/LTIFR/HIPOs, drives root-cause workflows

Category	Suggested Guardrail
Input validation	Validate incident report completeness (required fields, plausible dates/locations) before triggering a workflow
Output filtering	Root-cause summaries must avoid naming/blaming individuals without HSE review — focus on process/system factors
Tool/action permissions	Can open investigation workflows automatically; cannot close an investigation or determine final root cause without human sign-off
Human-in-the-loop	Final root-cause determination and corrective action plan require HSE Officer approval
Scope/topic boundaries	Limited to incident/investigation tracking; should not independently issue disciplinary recommendations
Rate limits / spend caps	N/A for volume, but cap duplicate workflow creation for the same incident
Output structure validation	Investigation record must follow the standard schema (TRIR/LTIFR fields, HIPO classification) for KPI rollups to work
Self-monitoring	Flag when investigation workflows exceed target closure time, indicating process bottlenecks
Fallback/escalation	If incident classification is ambiguous (recordable vs. non-recordable), route to HSE Officer rather than auto-classifying

10. Environmental Compliance Agent

Tracks CO₂/CH₄/NO_x/SO_x/waste/effluent vs. regulatory limits

Category	Suggested Guardrail
Input validation	Validate stack/effluent sensor data against calibration schedules — reject readings from overdue-calibration sensors
Output filtering	Rollup percentages/reports must state which monitoring points are included/excluded, not present a misleadingly clean aggregate
Tool/action permissions	Can track and flag ; cannot submit regulatory filings without human review (same principle as ESG Agent)
Human-in-the-loop	Environmental Lead reviews any trending-toward-breach flag before external communication

Scope/topic boundaries	Limited to environmental metrics; should not comment on safety incidents or financial impact
Rate limits / spend caps	Cap frequency of “trending toward breach” notifications per monitoring point
Output structure validation	Compliance rollup must match the regulator’s reporting schema for each pollutant category
Self-monitoring	Re-validate limit tables against the latest regulatory version periodically (regulations change)
Fallback/escalation	If a monitoring point goes offline, explicitly flag it as a gap rather than assuming compliance

11. Predictive Safety Intelligence Agent

Forecasts incident probability, heat stress, equipment failure 7-30 days out

Category	Suggested Guardrail
Input validation	Cross-check weather/roster/equipment inputs for staleness or gaps before generating a forecast
Output filtering	Forecasts must include a confidence interval and the underlying factors — never presented as a certainty
Tool/action permissions	Advisory only — flags high-risk zones; cannot alter shift rosters or equipment schedules directly
Human-in-the-loop	HSE Officer reviews flagged high-risk zones and decides on intervention (e.g., rescheduling, added monitoring)
Scope/topic boundaries	Forecasting limited to safety-relevant risk factors; should not be used for unrelated predictive HR decisions
Rate limits / spend caps	Cap the number of zones flagged simultaneously to avoid diluting attention (surface top-N by risk score)
Output structure validation	Forecast output must include zone, risk type, probability, and forecast window in a fixed schema
Self-monitoring	Track forecast-vs-actual incident correlation over time to detect model degradation
Fallback/escalation	If input data (e.g., weather feed) is unavailable, clearly mark the forecast as degraded/partial rather than omitting the caveat

12. Contractor & Asset Risk Agent

Aggregates contractor safety scores + asset failure risk into one site ranking

Category	Suggested Guardrail
Input validation	Validate contractor audit scores and asset condition data are current (not stale/expired audits) before including in ranking
Output filtering	Ranking outputs should avoid exposing individual contractor personnel data beyond aggregate scores; keep to organization-level scoring
Tool/action permissions	Can rank/flag ; cannot unilaterally suspend a contractor or take an asset offline — requires Plant Digital Head decision
Human-in-the-loop	Plant Digital Head reviews high-risk-flagged contractors before any contractual action (e.g., site access suspension)
Scope/topic boundaries	Confined to risk scoring/ranking; should not make procurement or contract-renewal recommendations directly
Rate limits / spend caps	Cap how often the ranking recalculates/broadcasts changes to avoid noisy reshuffling alerts
Output structure validation	Ranking output must follow a consistent schema (contractor/asset ID, score components, composite score)
Self-monitoring	Periodically audit whether score weightings still reflect actual incident/failure outcomes (recalibrate if not)
Fallback/escalation	If contractor audit data is missing, exclude from ranking with a visible “insufficient data” flag rather than defaulting to a score

Cross-cutting guardrails (apply platform-wide)

These extend what’s already in the Admin Console (human-in-the-loop for high-risk actions, explainability logging, model-change approval, data residency, quarterly red-teaming, PII masking):

- **Input validation:** A shared prompt-injection/anomaly filter for any agent that ingests free-text (incident notes, contractor comments) before it reaches an LLM reasoning step.
- **Scope/topic boundaries:** Agents should refuse or redirect questions clearly outside their domain (e.g., someone asking the Finance Agent for a safety opinion) rather than improvising an answer.
- **Rate limits:** A platform-level cap on total automated notifications per user per hour across *all* agents combined, to prevent cross-agent alert fatigue.
- **Fallback/escalation:** A shared “I’m uncertain” response pattern across agents (as seen in the chat fallback: *“I don’t have a specific*

answer for that yet...") so agents don't fabricate answers when the knowledge base has no match.